

THE CLEVER CODERS

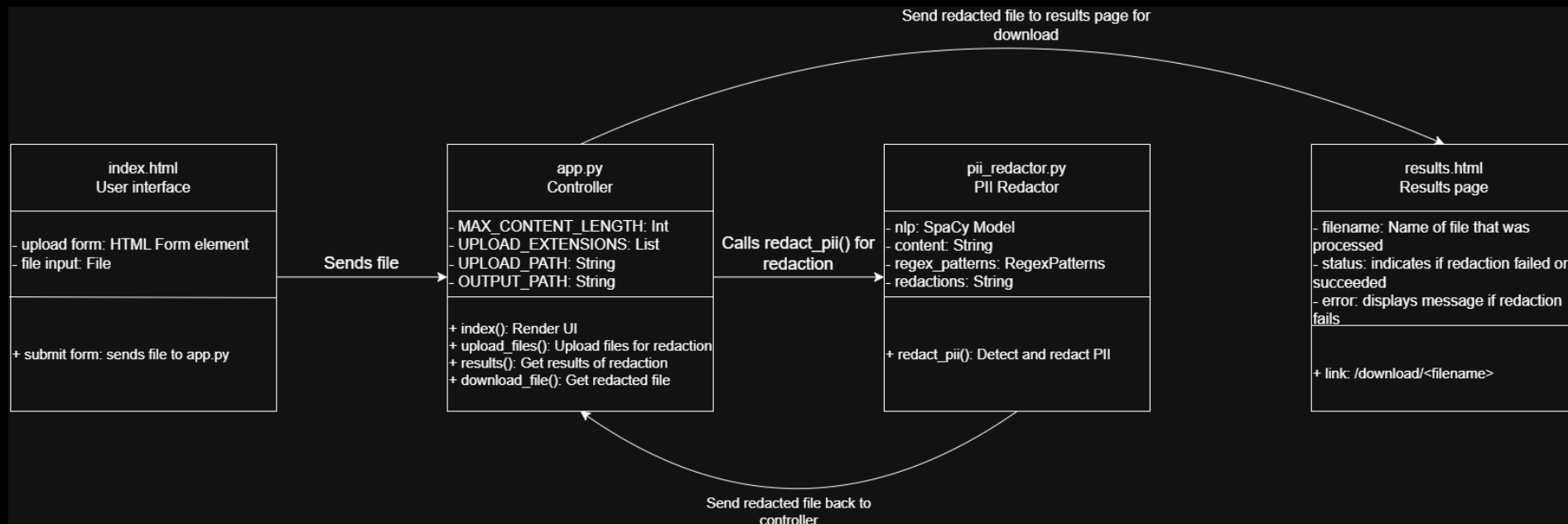
LOGAN LIVIGNI
ROBERT WELLS
SANG NGUYEN
ROCCO SWANEY
GAGE FLORES

Stage 2 Presentation



CLASS DIAGRAM

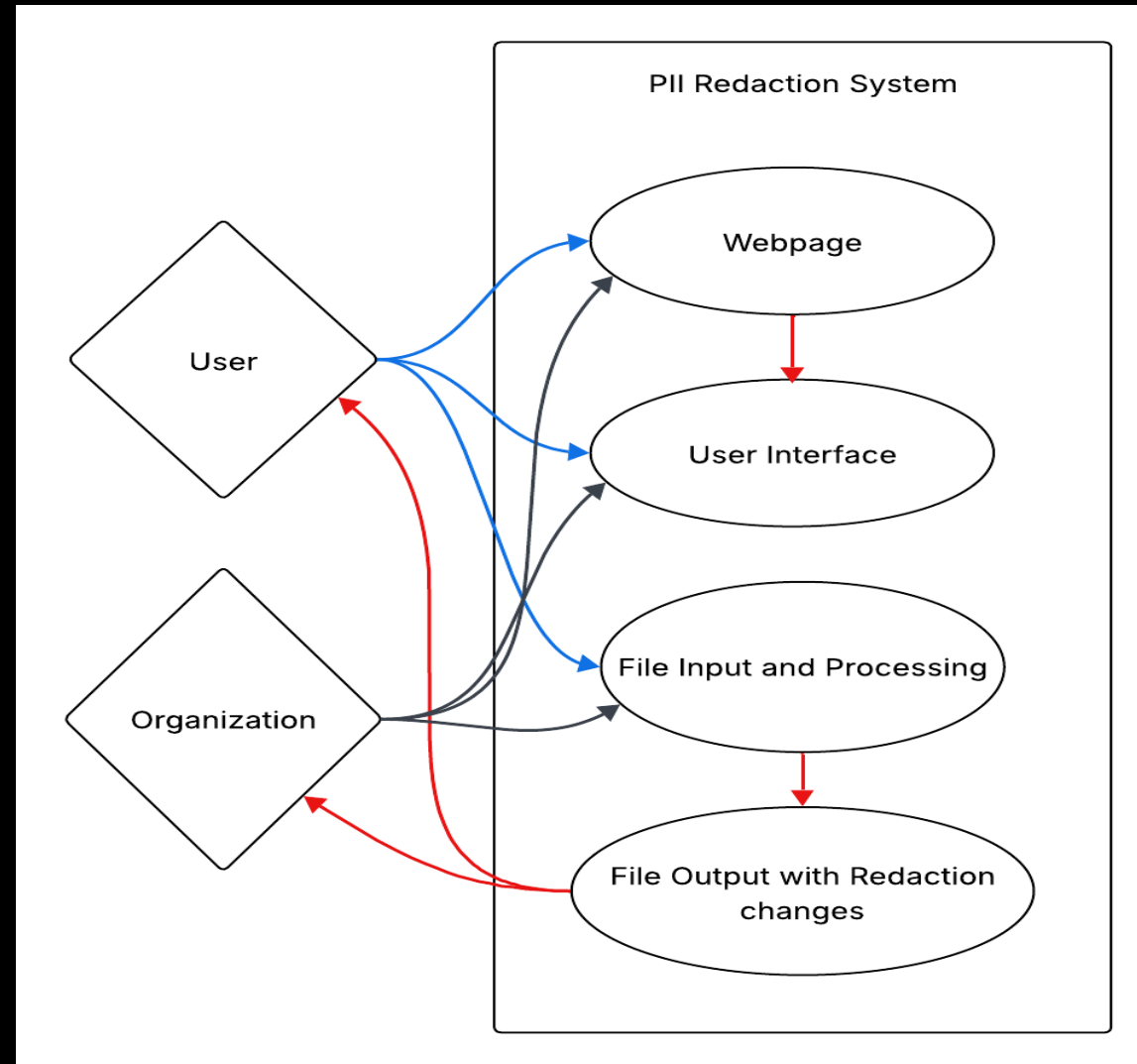
- 1. index.html – User interface for uploading file(s)
- 2. app.py – Controller that manages flow and calls redaction function
- 3. pii_redactor.py – Handles PII detection and redaction using SpaCy and regex
- 4. results.html – Displays redaction result and download link



USE CASE DIAGRAM

There are two main actors on the System, a User and an Organization.

While both have a similar purpose with the System the Organization will more often need to meet certain requirements for their document redaction, as they will have more legal legislation to abide by.

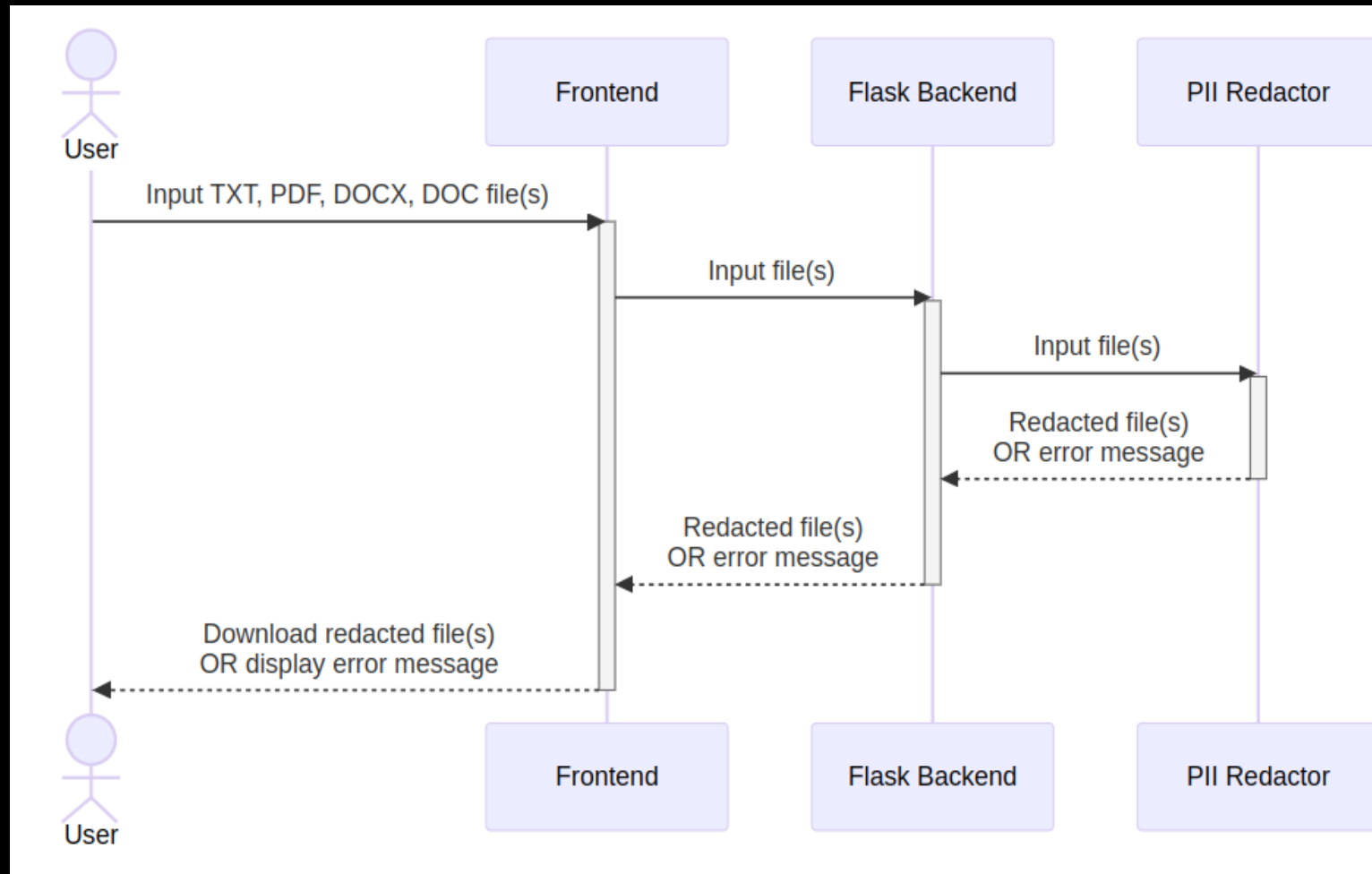


The PII Redaction System is broken down into 4 main uses.

1. Accessing the Webpage
2. Navigating the Webpage via the User Interface
3. File Input and Processing for desired redaction documents
4. File Output back to the user with redaction changes to their inputted documents

SEQUENCE DIAGRAM

1. A user will input their text document(s) on the website and submits it
2. Website sends documents to the backend
3. Backend executes Python script and passes the documents to it
4. Script returns redacted documents or error status to backend
5. Backend sends it to frontend
6. Frontend shows option to download files or display the errors



LOW-FIDELITY PROTOTYPE

PII Redactor

A simple tool that safely and instantly removes all personally identifiable information (PII) from your uploaded files.

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Smart Code. Clever Solutions.



Upload a file(s) (.txt, .pdf, .doc, .docx) to have the PII redacted.

Choose file No file chosen

Submit

Your file is processed securely and never stored.

TESTBED

- Testing will be done on local computers (Windows, macOS, Linux) to ensure cross-platform compatibility.
- Mock documents containing various PII (names, emails, phone numbers, SSNs, etc.) will be used during testing.
- The expected result is that the system identifies and redacts PII, replacing it with the correct placeholder without altering non-PII.
 - Ex: John Doe -> [NAME]

PROGRAMMING LANGUAGES

- **Backend:** Python
 - Chosen for its readability, portability, and extensive library support for text processing and pattern matching.
- **Frontend:** HTML/CSS
 - Chosen for its simplicity and effectiveness in building a UI. Combined with CSS for styling, HTML allows us to design the layout and user experience without complexity.

TOOLS & LIBRARIES

- **IDE:** Visual Studio Code or PyCharm - for coding and debugging
- **Version Control:** Git and GitHub - for source-code management and collaboration
- **Libraries:**
 - re – For regular expressions used to detect PII patterns.
 - os – For handling file operations.
 - SpaCy – For natural language processing.
- **Framework:** Flask - connects HTML frontend to Python backend